Venkata Lakshmi Parimala Pasupuleti

Software Engineer, Generative AI · AI Auto-Raters · LLM Evaluation · Agentic Systems · ML Infrastructure · Data Processing · Python · SQL · FastAPI · LangGraph

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PROFESSIONAL SUMMARY

Software Engineer with 8 years of production Python and data processing experience, and two shipped Generative AI systems in production. Built AgentProbe - an open-source AI auto-rater and ML infrastructure toolkit: LLM-as-judge scoring, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, and multi-provider LLM routing - the evaluation and quality assessment infrastructure for GenAI agents. Built LeaseI, an 8-agent Generative AI investigation platform (LangGraph + FastAPI) with conditional routing and HITL escalation. At Entrata, owned large-scale data processing pipelines (fragmented federal source data → 180,000 domain records), SQL query optimization (200+ queries, 30% performance gain), and scalable ML infrastructure for compliance-critical workflows at 99.5% uptime across 100K+ monthly active users. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, PL/SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, Linux, Git

Generative AI & Agents: Generative AI, GenAI, LLM Orchestration, AI Auto-Raters, LangGraph, LangChain, Agentic AI, Multi-Agent Systems, RAG Pipelines, ChromaDB, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Agent Observability, HITL, Prompt Engineering, Output Validation, OpenAI API, Anthropic API, MCP

ML & Data: ML Infrastructure, Model Evaluation, Data Processing, Feature Engineering, Optimization, PyTorch, scikit-learn, NLP, BERT, spaCy, PySpark, Apache Spark, ETL, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a Generative AI agentic mission-planning system (LangGraph stateful graph) with end-to-end ML infrastructure: parse → plan → retrieve → validate → conditional routing - LLM inference extracting structured flight plans from natural-language survey requests; validation guardrails enforcing FAA altitude limits, battery budgets, and no-fly-zone checks.
- Implemented RAG pipeline over FAA Part 107 corpus - every compliance check grounded in cited source regulation with LLM-based output validation and optimization for latency and accuracy.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$ regression accuracy; full ML pipeline from data processing and transformation through feature engineering, model optimization, and validation.
- Delivered Python and SQL workshops to 600+ students; designed Tableau and ggplot2 dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned large-scale data processing pipelines end-to-end for 7 years** - fragmented federal source data ingested, processed, and expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; designed validation and reconciliation at every stage; zero compliance violations.
- **SQL query optimization at scale** - optimized 200+ queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant schemas for eligibility, financial, and document workflows.
- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign - 50% scalability improvement, 30% deployment optimization; eliminated 80% of manual operational workload through automation.
- Built ML infrastructure for debugging and observability - internal tools reducing debugging time 40%; RabbitMQ async processing with retry logic and idempotency guarantees; AWS S3 migration with 100% PHI compliance (zero security incidents).
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; maintained bug count below 10 for 6+ consecutive months.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (PHP, Python, PostgreSQL); data processing pipeline for 1.6M HUD/Tax Credit records across 5 years; AWS S3 cloud migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI Auto-Rater & GenAI Evaluation Infrastructure Auto-Rater · ML Infra

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Open-source AI auto-rater and ML infrastructure toolkit for Generative AI evaluation: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge auto-rater scoring, pass@k harness assessing model headroom, CI-integrated regression gate blocking quality regressions before merge, and multi-provider LLM routing for cost optimization. Used to evaluate Leasel across 1,240+ compliance test cases.

Leasel: 8-Agent Generative AI Investigation Platform GenAI Agents

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Generative AI agentic system with 8 specialized agents (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation): LLM inference extracting root-cause hypotheses from fragmented data sources, conditional routing with HITL escalation when model confidence is too low. Achieved 93.6% precision, 87.9% F1 across 1,240 lease PDFs; CI regression gate with LLM-as-judge auto-rater scoring.

AI on FHIR: NLP-Driven Medical Records Query System LLM · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask + React NLP platform using LLM inference and BERT embeddings to translate natural-language queries into FHIR API calls - reduced manual clinical retrieval time ~60%; data processing across structured medical records with HIPAA-compliant RBAC.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017